Ahmed Khalil

Full-Stack Engineer (React · TypeScript · Java · Spring Boot) | Kassel, Germany

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Professional Summary

Web developer with 3+ years of experience building responsive, high-performance applications with React, TypeScript and WordPress. Solid full-stack capability with Java, Spring Boot, Spring Security (JWT) and PostgreSQL/MySQL, backed by an engineering background (M.Eng. Electrical Communication Engineering). Seeking a Junior Full-Stack Engineer role in Germany.

Technical Skills

- **Front-End:** React, TypeScript, JavaScript (ES6+), Next.js, Redux / Redux Toolkit, Context API, Tailwind CSS, HTML5/CSS3, Responsive Design, Performance Optimization (Lazy Loading, Intersection Observer)
- **Back-End:** Java (Core Java, OOP, Collections, Streams API), Spring Boot, Spring Data JPA / Hibernate, Spring Security, JWT Authentication, RESTful APIs, Maven, PHP, JSON Server
- **Databases & CMS:** PostgreSQL, MySQL, WordPress (Custom Plugins, REST API), Local Storage
- **Tools & Libraries:** Git/GitHub, Docker, VS Code, Leaflet.js, WebSockets, JavaFX, Geolocation API, Rate Limiting / Anti-Spam Protection

Work Experience

Web Developer / Freelance Software Engineer | Self-employed | 03/2022 – Present

- Designed and delivered responsive, accessible user interfaces in React, TypeScript and Tailwind CSS for more than 5 clients.
- Optimized web application performance using lazy loading, code-splitting, and asset optimization, improving initial load times.
- Developed custom WordPress plugins and REST API integrations tailored to client requirements.
- Integrated payment gateways (PayPal) with automated notifications and secure transaction handling.
- Built and deployed scalable web solutions end to end, from requirements through launch and maintenance.

Projects

Portfolio & Interactive AI Feature

Personal Developer Portfolio & AI Chatbot | Live

- Built a fully responsive portfolio with an AI-powered chatbot that answers visitor questions about my experience.
- Implemented rate limiting and anti-spam mechanisms to control API usage and protect the service.

Full-Stack & Systems Development (Java / Spring Boot & PHP)

Task & User Management System | React, TypeScript, Spring Boot, Spring Security, JWT, PostgreSQL, Docker

- Developed a full-stack application with REST APIs, secure JWT authentication, role-based access and CRUD for users and tasks.
- Containerised front end, API and database with Docker.

Inventory & Product Catalog API | React, Spring Boot, Spring Data JPA, MySQL

- Designed RESTful endpoints with data filtering and a React interface for managing the catalog.

Property Booking Platform (real-world project) | WordPress, PHP, Custom Plugins, REST API

- Built a booking platform with PayPal integration, real-time data processing, automated notifications and security measures.

2D Game Engine Application | Java, JavaFX

- Implemented an OOP architecture with animations and data persistence for high-score tracking.

Interactive Web Applications (React & JavaScript)

Real-Time Fitness & Location Tracker | JavaScript (OOP), Leaflet.js, Geolocation API, Local Storage

- Logged workout metrics and mapped routes in the browser with persistent storage.

Financial & Banking Operations Suite | JavaScript

- Simulated transfers, loan requests, interest calculations and secure session management.

Movie Library & Rating App | React, OMDB API, Custom Hooks, Local Storage

- Implemented dynamic search and state management with reusable custom hooks.

Real-Time Quiz & Learning Application | React, Context API, JSON Server

- Added a timer, dark mode and local score tracking.

Financial Marketing Site | JavaScript, Intersection Observer API

- Implemented lazy-loaded images, sticky navigation and smooth scrolling.

Education

Master of Engineering (M.Eng.), Electrical Communication Engineering | University of Kassel, Germany | 2026 – Present

Relevant coursework: Information Security, Cryptography Engineering, Advanced Mathematics, Signal/Data Processing

Bachelor's degree | Computer and Systems Engineering, Zagazig University, Egypt | 2020– 2025

Languages

English (Fluent) | German (A2.1) | Arabic (Native)